## Figure 3: Cross-Species Quantitative Divergence Matrix (n=54 Vertebrates)

| | α CDS GC (Z) | β CDS GC (Z) | Δ(α-β) CDS (Z) | α Gene GC (Z) | β Gene GC (Z) | Δ(α-β) Gene (Z) | Δ(α-β) Flank (Z) |
|---|---|---|---|---|---|---|---|
| Inshore Hagfish (E. burgeri) | +0.18 | | | +0.36 | | | |
| Sea Lamprey (P. marinus) | +0.43 | | | -0.07 | | | |
| River Lamprey (L. fluviatilis) | -0.36 | | | -0.03 | | | |
| Catshark (S. canicula) | +0.23 | -0.46 | +0.69 | -1.85 | -2.44 | +0.58 | |
| Spotted Gar (L. oculatus) | +1.09 | +1.22 | -0.13 | +1.02 | +1.09 | -0.07 | +0.00 |
| Zebrafish (D. rerio) | +0.61 | +0.39 | +0.23 | +1.91 | +1.03 | +0.88 | +0.55 |
| Minnow (P. promelas) | +0.24 | +0.47 | -0.22 | +0.88 | +1.90 | -1.02 | +0.00 |
| Medaka (O. latipes) | -0.43 | +0.47 | -0.90 | -0.16 | -1.28 | +1.12 | +0.80 |
| Fugu (T. rubripes) | +0.28 | -0.16 | +0.45 | +0.52 | -0.81 | +1.33 | +1.38 |
| Stickleback (G. aculeatus) | +0.00 | -0.12 | +0.12 | +0.67 | +0.92 | -0.25 | +0.00 |
| Gabon Caecilian (G. seraphini) | +0.29 | +0.38 | -0.09 | -0.24 | -1.06 | +0.82 | -0.43 |
| Two-lined Caecilian (R. bivittatum) | -0.45 | -0.27 | -0.18 | -1.56 | -1.00 | -0.56 | +0.00 |
| Tiny Caecilian (M. unicolor) | +0.70 | +0.74 | -0.05 | -0.55 | -0.54 | -0.01 | +0.00 |
| Axolotl (A. mexicanum) | +1.79 | +1.60 | +0.19 | +0.04 | +0.26 | -0.22 | +0.00 |
| Crested Newt (T. cristatus) | +1.91 | +1.39 | +0.52 | +0.09 | +0.21 | -0.12 | +0.99 |
| Clawed Frog (X. tropicalis) | +0.33 | +0.13 | +0.20 | -1.35 | -1.20 | -0.15 | +0.00 |
| Common Toad (B. bufo) | +0.60 | +0.85 | -0.25 | -1.26 | -0.98 | -0.28 | +0.00 |
| Painted Turtle (C. picta) | +0.54 | -0.64 | +1.18 | +1.07 | -0.41 | +1.48 | +0.64 |
| Sea Turtle (C. mydas) | +1.00 | +1.14 | -0.14 | +1.56 | +1.76 | -0.20 | -0.98 |
| Hawksbill Turtle (E. imbricata) | +0.93 | +1.04 | -0.11 | +1.57 | +1.52 | +0.05 | +0.00 |
| Softshell Turtle (P. sinensis) | +0.67 | +1.27 | -0.60 | +1.29 | +1.93 | -0.65 | -0.57 |
| Anole (A. carolinensis) | +1.24 | +1.69 | -0.45 | +2.76 | +2.79 | -0.02 | +2.60 |
| Bearded Dragon (P. vitticeps) | +1.63 | +1.54 | +0.09 | +4.49 | +4.18 | +0.31 | -0.06 |
| Wall Lizard (P. muralis) | +2.38 | +1.28 | +1.10 | +5.69 | +2.31 | +3.38 | +0.32 |
| Viviparous Lizard (Z. vivipara) | +1.76 | +1.61 | +0.15 | +4.50 | +3.38 | +1.12 | +0.98 |
| Ground Boa (C. aspera) | +1.07 | +1.48 | -0.41 | +2.17 | +3.25 | -1.08 | +0.90 |
| Adder (V. berus) | +0.69 | +1.97 | -1.27 | +1.80 | +3.80 | -2.00 | -0.09 |
| Tiger Snake (N. scutatus) | +0.73 | +2.05 | -1.32 | +2.21 | +4.09 | -1.88 | -0.38 |
| Corn Snake (P. guttatus) | +0.44 | +2.08 | -1.65 | +1.89 | +4.20 | -2.31 | +0.49 |
| Garter Snake (T. elegans) | +1.23 | +2.05 | -0.82 | +2.08 | +3.93 | -1.85 | +0.32 |
| Alligator (A. mississippiensis) | -0.21 | +0.43 | -0.64 | +0.20 | +0.56 | -0.36 | -0.35 |
| Ostrich (S. camelus) | +1.24 | +1.58 | -0.34 | +1.51 | +1.35 | +0.16 | -0.67 |
| Swan Goose (A. cygnoides) | +0.16 | +0.53 | -0.38 | +0.46 | +0.54 | -0.08 | +0.30 |
| Mallard Duck (A. platyrhynchos) | +0.24 | +0.67 | -0.43 | +0.40 | +0.92 | -0.52 | +0.00 |
| Turkey (M. gallopavo) | +0.66 | +1.03 | -0.37 | +0.84 | +1.25 | -0.41 | -0.30 |
| Quail (C. japonica) | +0.45 | +0.79 | -0.34 | +0.90 | +0.89 | +0.01 | +0.45 |
| Chicken (G. gallus) | +0.43 | +0.72 | -0.28 | +0.79 | +0.85 | -0.06 | +0.06 |
| Golden Eagle (A. chrysaetos) | +0.90 | +0.43 | +0.46 | +0.58 | +0.56 | +0.02 | +0.80 |
| Starling (S. vulgaris) | +0.66 | +0.92 | -0.26 | +0.95 | +1.46 | -0.51 | +0.06 |
| Platypus (O. anatinus) | +0.62 | +0.22 | +0.41 | +0.60 | +0.53 | +0.08 | +0.23 |
| Opossum (M. domestica) | +0.89 | +0.46 | +0.42 | +1.32 | +0.01 | +1.32 | +0.00 |
| Elephant (L. africana) | +1.17 | +0.08 | +1.09 | +2.08 | -0.33 | +2.41 | +0.00 |
| Mouse (M. musculus) | +0.70 | +0.02 | +0.67 | +1.59 | -0.45 | +2.04 | +1.82 |
| Norway Rat (R. norvegicus) | +0.70 | +0.09 | +0.61 | +1.49 | -0.57 | +2.06 | +0.47 |
| Naked Mole-Rat (H. glaber) | +0.68 | +0.04 | +0.64 | +1.19 | -0.05 | +1.25 | +0.56 |
| Bat (M. natalensis) | +1.12 | -0.21 | +1.33 | +1.86 | -0.60 | +2.46 | -0.08 |
| Puma (P. concolor) | +1.23 | +0.05 | +1.18 | +2.21 | -0.33 | +2.54 | +0.83 |
| Sea Lion (Z. californianus) | +1.21 | +0.14 | +1.06 | +2.10 | -0.33 | +2.43 | +0.00 |
| Pig (S. scrofa) | +1.40 | +0.36 | +1.04 | +2.16 | -0.07 | +2.23 | +1.36 |
| Cattle (B. taurus) | +1.21 | -0.17 | +1.37 | +2.13 | -0.50 | +2.63 | +1.40 |
| Dolphin (T. truncatus) | +1.04 | -0.07 | +1.10 | +2.03 | -0.28 | +2.31 | +0.00 |
| Donkey (E. asinus) | +1.26 | -0.06 | +1.32 | +2.39 | -0.43 | +2.82 | +1.68 |
| Horse (E. caballus) | +1.12 | -0.10 | +1.23 | +2.26 | -0.42 | +2.68 | +0.94 |
| Human (H. sapiens) | +1.46 | +0.12 | +1.34 | +2.75 | -0.42 | +3.17 | +1.38 |

